

Did you know...

Bitcoin has just over 67 digital currencies in circulation on the internet? See what they are: <http://dgit.in/bitcoin67>



Science meets Art

A theatre production: 'The Energy Show' at the Science Museum in South Kensington, London fuses science with theatre to inspire children <http://dgit.in/1eaHz68>

Work @ tech

is just another folder maintained exclusively by your operating system. When you delete the file from your recycle bin, it remains on your computer because the only thing the operating system deletes is the entry from the folder structure (the B-Tree Index); the data remains in place. This leaves the window to retrieve deleted data open. This is the window that data-recovery tools use. Undoubtedly, data retrieval in crime scene investigation is of higher importance than mere recovery of photos, thus the demand for highly skilled recruits.

- **Physical storage:** Data stored on a hard disk is stored as a sequence of bits, and each bit is physically stored by altering the magnetic value of a very tiny point on the disk. As imperfect as the world is, the magnetic value on the disk doesn't reflect a perfect binary value. For example, if the original state of a bit was at perfect 0, firing an electromagnetic pulse corresponding to 1 will cause the value to come to 0.92. Now, if you fire an electromagnetic signal to set that value to 0, it would come to a state that would probably indicate a value of 0.1 and then again you fire the pulse to set the bit to 1. This time, it goes to 0.95. Now, these values are fairly away from 0.5 and are easy to be determined as 1 or 0. However, if you notice the numbers above, a particular bit value can represent fair number of value overwrites on the same place. The accuracy of guessing the right value after x number of overwrites decreases as x increases. There are tools that do this fairly well. For one of these tools to work, they would need the exact magnetic value on the disk platter for each bit.



Mac OS X lets you remove the files in Trash securely without a third-party tool (just press Command key to show the option)

Not a lot of thought is required to conclude that to safely erase data from a disk. You'd need to overwrite the same disk portion where the data was stored, multiple times. The more random these values and more the number of overwrites, the lower the chance of correctly guessing the original value. The process of copying the bit-level detail of magnetic values on the disk is called "imaging". This it does by taking a snapshot of the entire disk/partition and saving it as a file. There are tools available for this process and a typical imaging session takes a lot of time because of the excruciating level of detail required. Speaking technically, imaging must always be done on the evidence-disk (the disk on which the evidence of crime is suspected to be present) in the read-only mode.

Information/Data interpretation

Once the disk image has been taken, the next task is to understand what the data means. This step involves many tools

and in-depth comprehension of the way different pieces of software interact with each other to trace back the circumstances of the felony. You may also be required to understand the operation of the OS and/or certain applications, which might have been used to commit the crime. While data interpretation is a complex process, we'll lightly touch upon it to give you an idea of the work involved in this step:

1. **Know the OS:** Data logged by operating systems sometimes plays a key role in determining what happened. Depending on the OS, the investigation areas might drastically change. Take Windows, for example; it stores a decent log of things that you try to do on the system. Activities such as installing an application and failed network login attempts are logged, and they can give a boost to an investigator's perception of what happened. Sometimes, one has to look into the registry to ascertain the tools that were used or their settings (many tools on Windows store their settings in registry).

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